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Design and experimental tests of a rotary active magnetic regenerator prototype

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Abstract

A rotary active magnetic regenerator (AMR) prototype with efficiency and compact design as focus points has been designed and built. The main objective is to demonstrate improved efficiency for rotary devices by reducing heat leaks from the environment and parasitic mechanical work losses while optimizing the utilization of the magnetized volume. Heat transfer calculations combined with 1D AMR modeling have revealed the necessity for an insulating air gap between magnet and regenerator when designing for high efficiency. 2D finite difference AMR modeling capturing the interplay between heat transfer fluid flow and an inhomogenous time-varying magnetic field in the individual regenerator beds has been used in the design process. For one operating point a COP of 3.1 at a temperature span of 10.2 K and a cooling power of 103 W were measured. Major issues limiting the performance have been identified and improvements are outlined for future work.

Keywords: Magnetic refrigeration; Design; Model; Efficiency; Experimental

1. Introduction

Magnetic refrigeration is a promising alternative to conventional vapor compression technology. It is based on the magnetocaloric effect in ferromagnetic materials, hereafter referred to as magnetocaloric materials (MCM). As a consequence of this effect, the temperature of an MCM will, under adiabatic conditions, change as a response to a change in an applied magnetic field, such that the temperature will increase when the field is increased and vice versa. The effect, which is most pronounced in the vicinity of the Curie temperature of the material, can be utilized in heating or cooling devices; see e.g. Smith *et al.* (2012). In 1982 Barclay demonstrated a cooling device based on a concept where the MCM itself was used to regenerate heat which was transported between a cold and a hot reservoir via a heat transfer fluid; see Barclay, J. A. (1982). Since then this principle, known as the active magnetic regenerator (AMR) cycle, has been used in an increasing number of devices with the aim of making magnetic refrigeration near room temperature a competitive alternative to conventional vapor compression technology. For a more comprehensive description of the AMR cycle; see Engelbrecht *et al.* (2012). In general, magnetic refrigeration has the advantage of not using gaseous refrigerants. The absence of a compressor opens the possibility of silent operation. Furthermore, some of the losses associated with vapor compression are avoided, which may lead to a higher efficiency. Already in 1998, COPs above 6 were obtained at cooling powers exceeding 500 W, see Zimm *et al.* (1998). However, this device used a liquid helium cooled superconducting 5 T magnet and the power used for the magnet itself was not included in the COP calculation. Superconducting magnet AMRs are not economically viable with the present technology. Over the years, permanent magnets have been used in an increasing number of published devices; see e.g. Bjørk *et al.* (2010a) and it seems that devices based on rotary concepts with permanent magnets have a good potential for high performance; see e.g. Yu *et al.* (2010).

A design with a rotary magnet structure and regenerator with reciprocating flow provided by a displacer has produced high temperature spans, e.g. a temperature span of 29 K has been achieved by Tura and Rowe (2011). Recently, a temperature span of 33 K was demonstrated for an improved version of the same device; see Arnold *et al.* (2014). In other devices, a compartmentalized regenerator and a magnet system are mechanically rotated relative to each other. The flow system in these devices is of major importance. By applying

valve systems to control the heat transfer fluid flow, it is possible to achieve a continuous flow circuit driven by a pump to transport heat to and from the regenerator while ensuring a reciprocating flow in the regenerator compartments; see eg. Tusek *et al.* (2010), Engelbrecht *et al.* (2012) and Jacobs *et al.* (2014). By carefully balancing magnetic forces and fluid flow, a smooth and efficient operation may be obtained. However, during each AMR cycle, the work performed on the regenerator is negative during magnetization and positive during demagnetization. In order to obtain efficient operation which is comparable to numerical AMR model predictions, the work performed by the regenerator during magnetization has to be utilized. In a multi bed rotary AMR configuration, such as the one presented in this paper and the earlier device presented by Engelbrecht *et al.* (2012), this is done by always having beds moving into the magnetic field, thus contributing to the driving torque, while others are moving out. However, a poorly mechanically and magnetically balanced configuration will result in large torque fluctuations and drivetrain losses. How to achieve an optimum configuration in this regard is not well understood. With the presented device, an improved drivechain is realized by minimizing the thickness of the walls separating the regenerator beds in order to obtain a more even distribution of MCM and hence a more smooth driving torque. Furthermore, an odd number of regenerator beds are combined with a two pole magnet in order to avoid magnetic equilibrium positions during rotation.

Although much work has been conducted over the years to improve the efficiency of the devices, there are still significant technical challenges that need to be overcome. Recently, it was shown that care should be taken to reduce a number of parasitic losses associated with the overall COP of an AMR system; see Lozano *et al.* (2013). In this paper we report a compact new regenerator design, bringing these losses down. In the process, detailed numerical AMR modelling has been used as a design tool to address heat loss issues and the inhomogeneous magnetic field in regenerator compartments.

2. System design

The AMR device consisting of a regenerator, a magnet and a flow control system is shown in Figure 1. For the device presented here, the MCM is confined in a cylindrical regenerator which is divided into eleven compartments. The regenerator is fixed on the outside of an iron core that is a part of the magnet system. The core consists of laminated plates of iron and glass fiber

reinforced epoxy (GFRE) in order to minimize losses due to eddy currents and thermal conduction in the axial direction. On the upper and lower sides of the regenerator, valve arrangements ensure reciprocating flow of the heat transfer fluid in the regenerator compartments and a continuous, unidirectional flow in the external flow circuit. Special care has been taken to reduce losses on the cold lower side of the regenerator. The outer magnet ring, which is the only rotating part of the device, is situated on the outside of the regenerator and mounted concentrically with the iron core. The structure is rotated at an AMR frequency which can be varied between zero to approximately 4 Hz. The rotating outer part not being mechanically connected to a central shaft enables simple connection between the flow distribution system mounted above the regenerator and the external flow circuit. On the hot top side of the device, hoses supply fluid in and out of 11 regenerator beds timed with the rotation of the magnet. The timing of the flow is ensured by poppet valves in the flow distributor. These valves are actuated by cam rings rotating along with the magnet. A detailed description of the flow system will be available in Eriksen *et al.* (2015). The excess heat in the hotter outgoing fluid is rejected in the hot heat exchanger (HHEX), before going back into the regenerator. On the cold side, the fluid goes through an electrical heater, simulating a heat load.

2.1. Magnet system

A cross section of the magnet, iron core and regenerator is shown in Figure 2a. The magnet consists of a number of blocks of permanent NdFeB magnets, magnetized in the indicated directions. In combination with flux-conducting iron yokes and an iron core, the structure forms a Halbach-like array with two high field and two low field regions in the gap where the regenerator is situated. These regions will be swept over the regenerator as the magnet rotates.

The 12 pieces of permanent NdFeB magnet, grade N50, have a total volume of 1.5 L. The flux density in the gap between iron core and magnet was measured using a three-axis Hall probe (Arepoc s. r. o. AXIS-3) as a function of radius, length and azimuthal angle. There is a good agreement between modeled and measured flux density; see Figure 2. However, modeling has also shown that a higher flux density may be obtained if the insulation between the iron discs in the core is reduced to a minimum. The calculations reveal a potential to increase the maximum flux density from the current 1.13 T to 1.4 T. This is considered for future work.

2.2. Regenerator

A photograph showing the regenerator filled with spheres of Gd and three different compositions of $\text{Gd}_{100-x}\text{Y}_x$ from Santoku Corp. (Japan) is shown in Figure 3. These materials are arranged such that they will operate near their respective Curie temperatures. The diameter of the spheres is between 500 μm and 600 μm for the Gd and between 300 μm and 500 μm for the $\text{Gd}_{100-x}\text{Y}_x$ compounds. As the magnitude of the magnetic flux density decreases near the ends of the 100 mm long magnet, only the central 90 mm is utilized as AMR. The total mass of MCM is 1.7 kg. The regenerator housing consists of two concentric non-magnetic stainless steel cylinders, grade 304, with a thickness of 0.5 mm. The walls separating the 11 regenerator beds are made from 0.5 mm thick GFRE plates. This material was chosen to avoid eddy currents while ensuring sufficient mechanical stability.

After packing the regenerator with MCM spheres, it was closed at the ends with PVC/PA parts with stainless steel wire mesh screens facing the MCM and holes for flow system connections. The dimensions of the housing material inside the magnet were minimized using the commercially available finite element software Comsol Multiphysics for strength calculations, taking expected temperature and pressure gradients into account. This was done to optimize the volume available for MCM in the magnetized volume. Furthermore, minimizing the thickness of the walls separating the regenerator compartments gives a more consistent magnetization of the regenerator and enables a smooth driving torque as the magnet is rotated.

3. Modeling used as design tool

This section describes the use of numerical modeling as a tool for addressing both heat losses and the effects of having an inhomogeneous magnetic field in the regenerator compartments. The modeling has been used for the design and dimensioning of the presented prototype. However, the issues addressed are inherent to the considered type of rotary devices.

The modeling is based on a 1D numerical AMR model solving coupled governing equations based on energy balances for the MCM and heat transfer fluid respectively. The model takes property data for these materials, given fluid flow and applied magnetic field as functions of time during the cycle and a given temperature span and cycle frequency as inputs. The main outputs are the hot and cold heat flows and the input power. The model was originally validated experimentally against an AMR prototype described by

Zimm *et al.* (2006), see Engelbrecht (2008). The model that is used to capture spatial variations in magnetization in a regenerator bed solves the same governing equations in 2D as described below. This model was based on previous passive 2D porous bed regenerator modelling described in Nielsen *et al.* (2013).

3.1. Parasitic heat loss through regenerator housing

Special attention has been given to the heat leakage between the regenerator and the magnet/iron core as earlier work has identified this as an important issue limiting the system performance; see Lozano *et al.* (2013). The situation is illustrated in Figure 4. Between the regenerator housing and the rotating outer magnet is an air gap. The magnet, having a large thermal mass and good thermal transport properties, is considered to be at the ambient temperature, T_∞ , and the regenerator wall is considered to have a linear temperature gradient between T_C and T_H in the z -direction. The heat transfer through the regenerator housing due to both forced and natural convection has been considered. The rotation of the magnet will create a friction driven air flow in the gap, which is treated as a Couette flow; see e.g. Incropera and DeWitt (1996). This simple case of forced convection has an analytical solution revealing that the heat flux in the radial direction consists of only two terms: viscous dissipation in the air and conduction driven by the temperature difference between regenerator and magnet.

For the rotational speeds relevant for the present application, it is found that the viscous dissipation may be neglected. Furthermore, the natural convection was modeled with a simple finite element model made in Comsol Multiphysics. Based on this it has been concluded, that for an air gap of up to a few mm, the heat transfer is still dominated by conduction. Furthermore, the natural convection may be limited by physical barriers to the axial flow. Therefore, only the heat transfer by conduction is considered in the following, both for the gap between regenerator and rotating outer magnet and for the equally wide gap between gap between regenerator and iron core. Since the present device design was focused on efficiency, estimates of the influence of how the conduction based loss influences the COP have been made, based on the geometry shown in Figure 4, with $d = 100$ mm, $D = 132$ mm and $L = 90$ mm. It was assumed that the regenerator housing consisted of stainless steel with a thickness of 0.5 mm. For simplicity, it was assumed that the hot end of the regenerator was at the ambient temperature of 30°C and that the cold end was at 0°C . In this situation, all the heat flux is

going into the regenerator from the magnet and the iron core, which was not considered to be laminated in this case. Then, the total heat flux into the inside and outside of the regenerator was integrated over its length, giving a total heat loss, $\dot{Q}_{Loss}$. The cooling power, $\dot{Q}$, produced by the device and the work input, $\dot{W}$, which consists of both magnetic work, $\dot{W}_{mag}$ and pump work, $\dot{W}_{pump}$, i.e.

$$\dot{W} = \dot{W}_{mag} + \dot{W}_{pump}, \quad (1)$$

was calculated using the 1D numerical AMR model described above, see also Engelbrecht and Bahl (2010), assuming a varying flow rate and an AMR operational frequency of 0.75 Hz. The COP was then evaluated as:

$$COP = \frac{\dot{Q}}{\dot{W}}. \quad (2)$$

Next, as a rough estimate, the calculated loss was simply subtracted from the cooling power when calculating the COP with losses, COP_L :

$$COP_L = \frac{\dot{Q} - \dot{Q}_{Loss}}{\dot{W}}. \quad (3)$$

Results from these calculations are shown in Figure 5a, where the flow rate is increased from 4.2 L min^{-1} , corresponding to a utilization of $\phi = 0.4$, until the cooling power reaches zero. Here, the utilization is defined as

$$\phi = \frac{m_f c_f}{m_s c_s}, \quad (4)$$

where m_f is the mass of fluid flowing through a bed during a blow period, m_s is the mass of MCM in a bed and $c_f = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$ and $c_s = 300 \text{ J kg}^{-1} \text{ K}^{-1}$ are the heat capacities of the fluid and MCM respectively. In each case, the cooling powers plotted in Figure 5a reach their maximum values at approximately $\phi = 0.8$. It is clear that, without the loss term included, the regenerators with the smaller air gaps and hence higher masses perform better in most cases, both in terms of cooling power and COP. However, when the utilizations become low, higher COPs can be achieved with larger air gaps. This is considered to be an effect of smaller air gaps giving larger cross sectional areas, which leads to increased axial heat conduction losses. This is captured directly by the AMR model. The figure furthermore shows, that when the heat losses through the regenerator walls are included, the COPs of the smaller air gaps become significantly decreased. Therefore, when

designing for high COP, it will be desirable to have a certain air gap, which in this case is larger than 1 mm, even though it slightly decreases the maximum cooling power. In Figure 5b the COP is plotted as a function of the air gap for different cooling powers. Based on these considerations, an air gap of 2 mm was chosen in the present case. Such a tradeoff between insulating air gap and the amount of MCM in the magnet is inherent to this type of rotary AMR devices. Having a clearance between rotating and stationary components furthermore has the advantage of reducing the demands for manufacturing tolerances.

3.2. Inhomogeneous magnetic field and 2D AMR modeling

The field in a regenerator bed is inhomogeneous, meaning that the MCM experiences an applied field that varies with the azimuthal direction. This makes it non-trivial to optimize the flow rate of the heat transfer fluid during the cycle because the entire bed experiences the same fluid flow rate. To address this issue, a finite difference AMR model, based on previous work by Nielsen *et al.* (2013), was developed that resolves the regenerator in both the z and ϕ directions indicated in Figure 2. This 2D model has been used as a design tool to iterate between different combinations of magnet design, regenerator design and flow profile (flow rate versus rotation) design. In this process, different magnetic field profiles have been developed and investigated, using the iterative finite element based methodology described in Bjørk *et al.* (2010b) and Bjørk *et al.* (2011). This has been done in combination with a variety of regenerator and flow profile designs modeled with the 2D AMR model. The investigated flow profiles are trapezoidal, meaning that they consist of full flow periods, no flow periods and linear ramps. This combined modeling approach has been used as a tool to design the present prototype and the methodology will be described in further detail in future works.

Once the inner and outer dimensions of the magnet and regenerator were fixed, the number of beds was varied in the 2D AMR model; see Figure 6. It is a clear result that having more beds is desirable, although not from a practical standpoint because many beds require smaller dimensions, more valves and more intricate assembly. The number of beds was chosen to be eleven as a compromise in the present case. This number was also chosen not to be a multiple of the number of magnetic poles to improve mechanical efficiency.

4. Interfacing

Adjustable operational parameters and instrumental readings are handled by a dedicated LabView PC program, allowing the control of operational frequency and heater power. Data were acquired using a National Instruments system (cDAQ-971). The fluid flow rate is regulated manually by needle valves. The hot side temperature is controlled by a recirculating cooler (Julabo FL4003) connected to the hot counter flow heat exchanger. The inlet and outlet temperatures on the hot and cold sides of the regenerators are measured by type K thermocouples and the fluid flow rate is measured by a magnetic inductive flow sensor (Omega FMG71). Furthermore, a portable flowmeter (Omega Microflow FTB322D) can be inserted on the connecting hoses indicated in Figure 1 to measure the actual flow in any of the eleven regenerator beds.

5. Experimental results

AMR experiments have been conducted with the hot side temperature held constant around 18°C. A series of experiments at an AMR frequency of 0.75 Hz and a fluid flow rate of 3 L min⁻¹ resulted in the cooling curve shown in Figure 7. The indicated COP values are calculated according to Eqs. 1 and 2 with $\dot{Q}$ being the power supplied to the heater providing the cooling load. $\dot{W}_{mag}$ is taken as the power supplied to the motor and $\dot{W}_{pump}$ is evaluated as the measured fluid flow rate times the total pressure drop over the regenerator. Thus, the efficiency of the pump itself is the only parameter which was not included in the COP. The current pump is over dimensioned in order to be able to cover a wide range of pressure drops and flow rates for experimental purposes. Demineralized water with 5% ethylene glycol based automotive antifreeze was used as heat transfer fluid. Measured as a fraction of the Carnot efficiency, the best result is obtained at a temperature span of 10.2 K at a cooling load of 102.8 W. In this case the COP of 3.1 is 11.3% of the Carnot Efficiency. The highest COP values obtained so far at different temperatures are plotted in Figure 8, where a significant improvement compared to the first rotary DTU prototype is also shown, see Lozano *et al.* (2013). Beside the minimized parasitic losses, the driving torque fluctuations are greatly reduced, hence drive train losses are also reduced as described above. So far the device has only been operated at very modest conditions, especially in terms of flow rate and pressure drop, in order to

avoid possible mechanical overloading of the regenerator.

The actual flow rates in some of the regenerator compartments have been measured as a function of magnet rotation angle using the portable flow meter. The result of such a measurement can be seen in Figure 9, where also the expected flow profile is indicated together with the magnet profile. It is clear, that there is a significant discrepancy between the expected and measured flow rates, indicating that adjustments to the flow system are needed. Already, the system has shown strong responses to such adjustments. It is evident that, for the bed measured in this case, the total flow is less in the cold to hot direction than in the hot to cold direction. This is hypothesized by the authors to be due to the way different beds with somewhat different flow resistances are connected during operation. This flow balancing issue might also be an important factor limiting the achieved temperature spans, which are quite modest compared to what could be expected from the present staggered regenerator using materials with four different Curie temperatures as described in the introduction.

The potential for getting a higher magnetic field with a different iron core as mentioned above combined with adjustments to the flow system indicate a good potential for improved results in the future.

6. Conclusion

A rotary AMR prototype has been designed and built. The focus has been on enhanced performance in terms of efficiency and compact design at temperature spans and cooling powers relevant for commercial applications. During the design process, heat transfer calculations combined with 1D AMR modeling have revealed that there will be a trade-off between the amount of MCM and insulating air gap between magnet and regenerator in rotary prototypes when designing for high COP. Furthermore, 2D AMR modelling has proven to be an important tool for the design of AMR device configurations where the magnetic field varies as a function of rotation angle. Experimental results from the presented prototype have shown promising results, including a temperature span of 10.2 K at a cooling load of 103 W and a COP of 3.1. A potential for obtaining a higher magnetic field and the need for improving the balancing of heat transfer fluid flow have been identified and significant improvements of system performance are expected in the future.

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Figure 1: (a) Photo and overall dimensions of the AMR device.(b) Simplified cross section and diagram with external components.

Figure 2: (a) Rotating magnet system consisting of 12 NdFeB blocks connected by iron yokes. In the middle: stationary iron core and regenerator in the gap. (b) Measured and modeled field.

Figure 3: Regenerator divided into 11 compartments, filled with spheres of Gd and different compositions of $Gd_{100-x}Y_x$. The drawing shows the Curie temperature of each layer in a compartment. All sizes are in mm.

Figure 4: (a) Regenerator with temperature gradient situated between iron core and rotating outer magnet. (b) Air gap between regenerator and rotating magnet at a given z . Forced convection with linear velocity profile between 0 and U and heat flux, q'' , into the regenerator.

Figure 5: (a) Modeled COP and cooling power with varying flow rate. Different thicknesses of air gap, with and without losses included in the COP calculation. (b) Highest COP from the same data set at fixed cooling powers plotted as a function of air gap.

Figure 6: Cooling power and COP versus number of beds and resulting width of the individual regenerator compartments. The utilization is 0.4

Figure 7: Cooling curve with corresponding COP values. The utilization was 0.3

Figure 8: COP versus temperature span showing significant improvement compared to rotary prototype previously presented by DTU.

Figure 9: Normalized magnet and measured flow profiles in one of the eleven regenerator compartments. Red indicates hot to cold and blue indicates cold to hot blow.

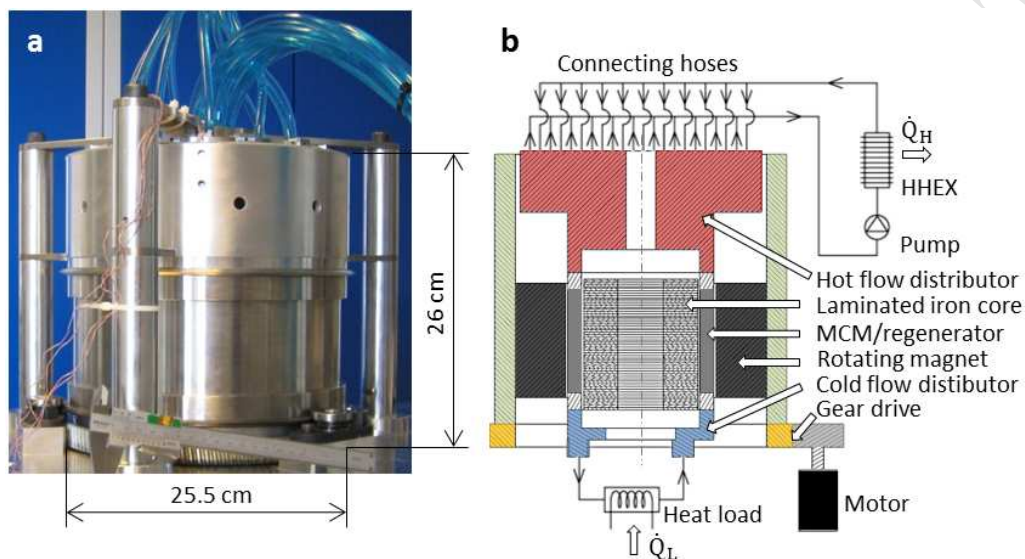


Figure 1

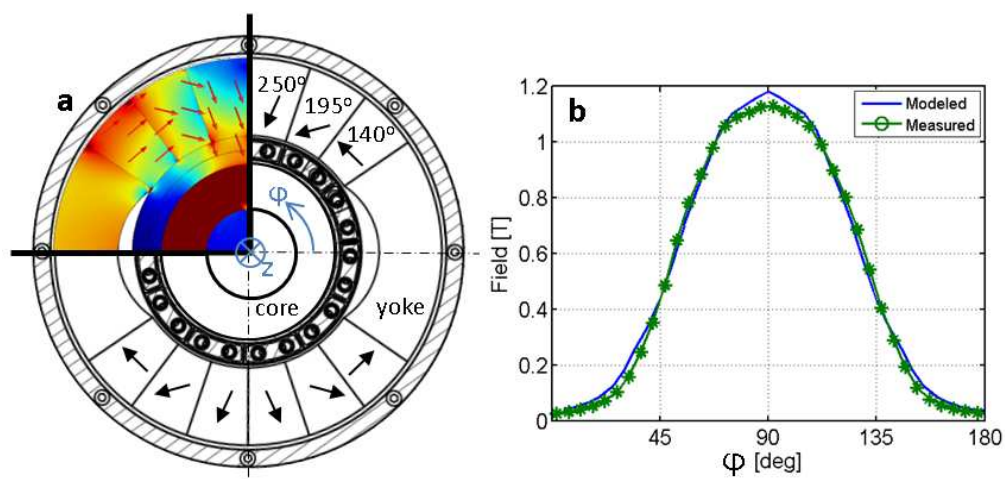


Figure 2

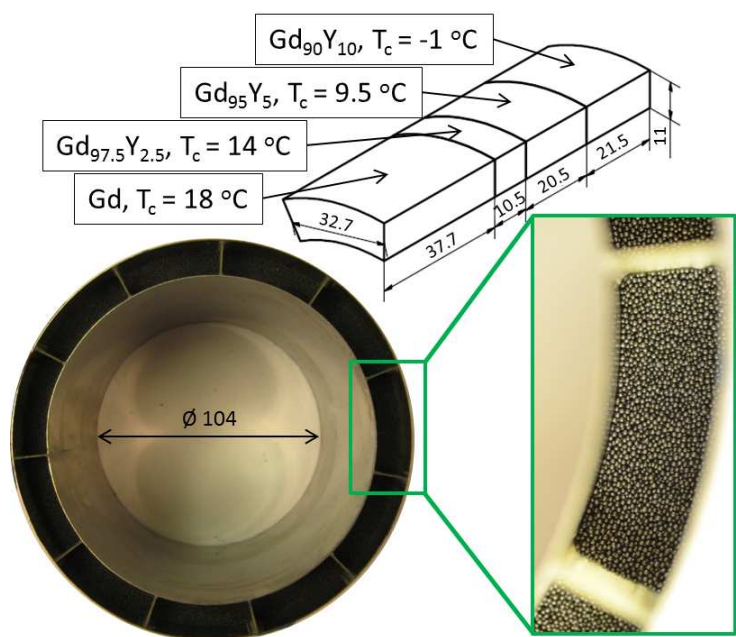


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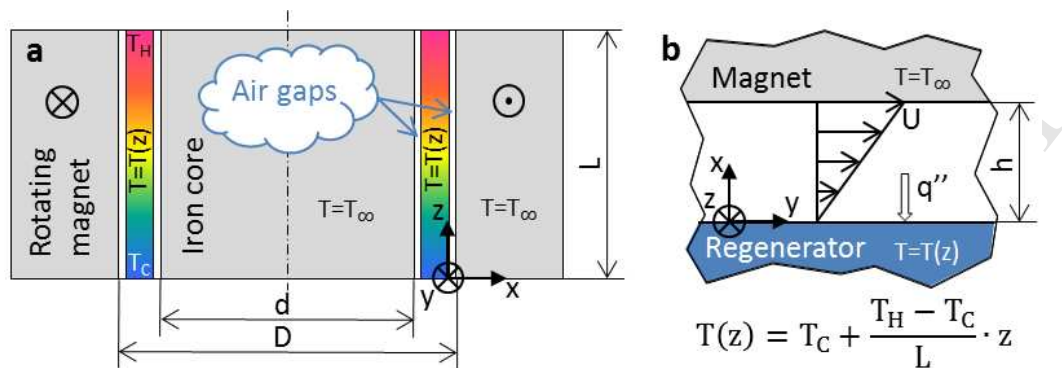


Figure 4

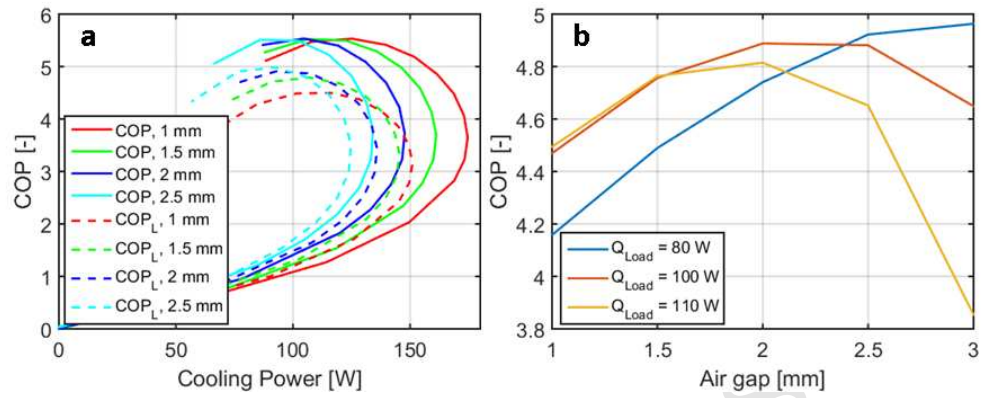


Figure 5

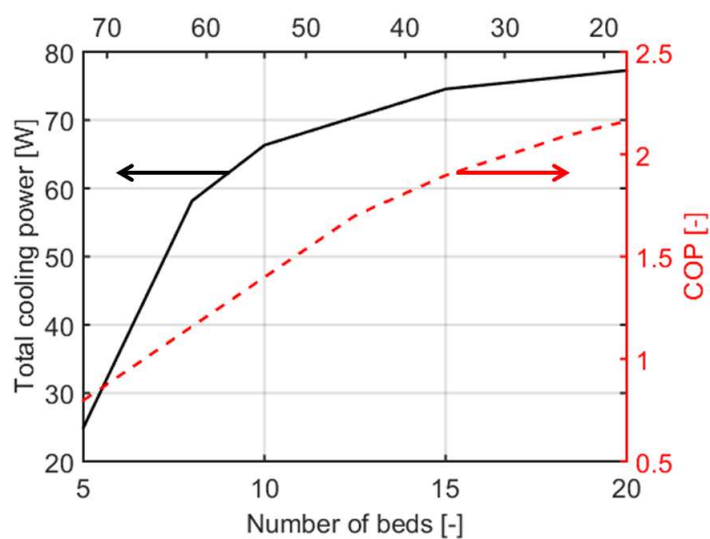


Figure 6

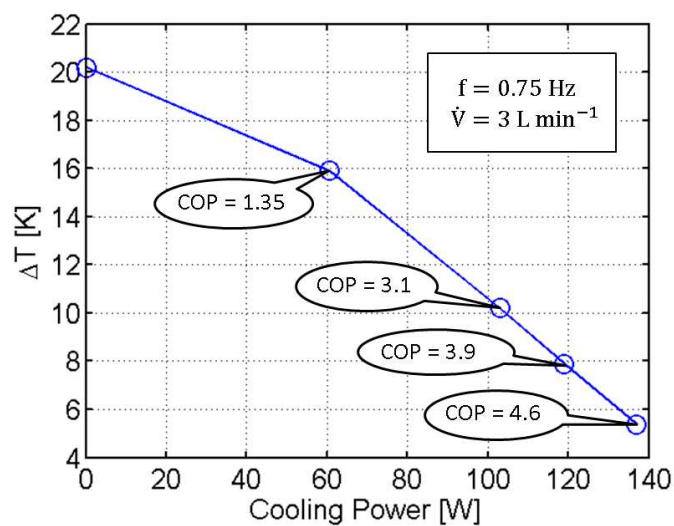


Figure 7

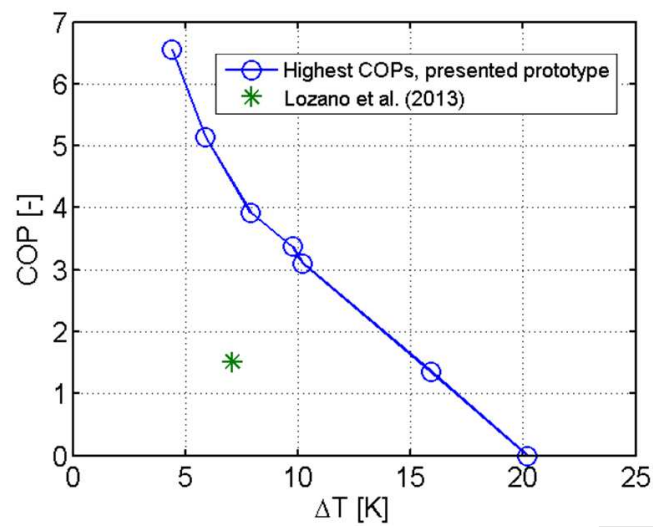


Figure 8

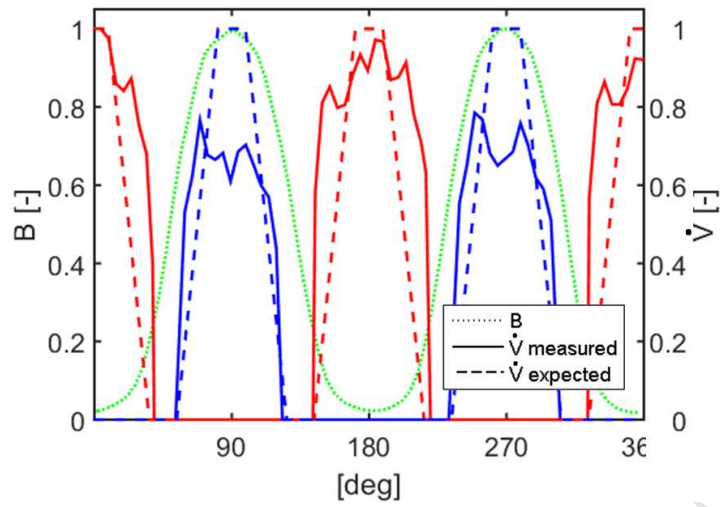


Figure 9

Highlights

- Rotary active magnetic regenerator prototype was designed, built and tested
- Modeling reveals a benefit from an insulating air gap between regenerator and magnet
- Modeling capturing flow/inhomogeneous magnetic field interplay was used for design
- COP of 3.1 was achieved at a temperature span of 10.2 K and a cooling power of 103 W